

Date: _____

Assignment No 45

Day: _____

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Subject: Programming Fundamentals

Topic: "Formal, Actual, Parameter, Local, Global, Inline Function"

Theory Paragraph:

variables are function play an essential role in C++ programming. variables are used to store and manage data, while function help break a program into a smaller part, making it easier to understand, test and maintain. Formal parameters are the variables defined in a function, global variables are declared outside all functions and remain available throughout program's execution.

Code Examples

Example 1: Formal and Actual Parameters

```
#include <iostream>
```



```
'Using namespace std;
void add(int a, int b) // a and b
are formal parameters
{
```

```
    int sum = a + b;
    cout << "Sum = " << endl;
}
```

```
int main () {
    int x = 5, y = 10; // actual parameters
    add(x, y);
    return 0;
}
```

out put :- Sum = 15

Explanation:-

a and b formal parameters
x and y are actual parameters passed
to the function.

Example 2: Local and Global variables

```
#include <iostream>
```

```
using namespace std;
```

```
int globalVar = 100; // Global variable
```

```
void display ()
{
```



```
int localvar=50; // local variable  
cout<<"local variable="<<localvar  
<<endl;
```

```
}
```

```
int main() {
```

```
display();
```

```
cout<<"Global variable="<<globalvar  
<<endl;
```

```
return 0;
```

```
}
```

Output :- Local variable = 50

Global variable = 100

Explanation Localvar can only be used inside the display() function.

- globalvar can be accessed anywhere in the program.

Example 3: Inline Function

```
#include <iostream>
```

```
using namespace std;
```

```
inline int Square(int x)
```

```
{
```

```
return x*x;
```

```
}
```


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```
int main () {  
    int num=4;  
    cout<<" Square = "<<Square (num)  
    <<endl;  
    return 0;  
}
```

Output: Square 16

Explanation: The inline function reduce function call overhead and improves performance for small function.